

1.

```
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> gcc hello world.c -o hello world  
  
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> ./hello_world  
Hello World! 🌍
```

2. С использованием if else:

```
~/University/Innoplis/Programming for gamedeveloping/Homework/Homework 1 master*  
> gcc fsm_if_else.c -o fsm_if_else  
  
~/University/Innoplis/Programming for gamedeveloping/Homework/Homework 1 master*  
> ./fsm_if_else  
Input symbol: a, Start state: 0, End state: 1  
Input symbol: b, Start state: 1, End state: 3  
Input symbol: c, Start state: 3, End state: 1  
Input symbol: d, Start state: 1, End state: 0  
Input symbol: a, Start state: 0, End state: 1  
Input symbol: b, Start state: 1, End state: 3  
Input symbol: c, Start state: 3, End state: 1  
Input symbol: d, Start state: 1, End state: 0  
Input symbol: a, Start state: 0, End state: 1  
Input symbol: b, Start state: 1, End state: 3  
Input symbol: c, Start state: 3, End state: 1  
Input symbol: d, Start state: 1, End state: 0  
Input symbol: a, Start state: 0, End state: 1  
Input symbol: b, Start state: 1, End state: 3  
Input symbol: c, Start state: 3, End state: 1  
Input symbol: d, Start state: 1, End state: 0  
  
Input symbol: a, Start state: 0, End state: 1  
Input symbol: a, Start state: 1, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
Input symbol: b, Start state: 2, End state: 3  
Input symbol: b, Start state: 3, End state: 3  
Input symbol: b, Start state: 3, End state: 3  
Input symbol: b, Start state: 3, End state: 3  
Input symbol: c, Start state: 3, End state: 1  
Input symbol: c, Start state: 1, End state: 1  
Input symbol: c, Start state: 1, End state: 1  
Input symbol: c, Start state: 1, End state: 1  
Input symbol: d, Start state: 1, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: a, Start state: 0, End state: 1  
Input symbol: a, Start state: 1, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: c, Start state: 0, End state: 3  
Input symbol: c, Start state: 3, End state: 1  
Input symbol: c, Start state: 1, End state: 1  
Input symbol: c, Start state: 1, End state: 1  
Input symbol: b, Start state: 1, End state: 3  
Input symbol: b, Start state: 3, End state: 3  
Input symbol: b, Start state: 3, End state: 3  
Input symbol: b, Start state: 3, End state: 3  
Input symbol: a, Start state: 3, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
Input symbol: a, Start state: 2, End state: 2  
Input symbol: d, Start state: 2, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0  
Input symbol: d, Start state: 0, End state: 0
```

С использованием switch case:

```
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> gcc fsm_switch_case.c -o fsm_switch_case  
  
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> ./fsm_switch_case  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: b, Start state: 1, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: d, Start state: 1, End_state: 0  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: b, Start state: 1, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: d, Start state: 1, End_state: 0  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: b, Start state: 1, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: d, Start state: 1, End_state: 0  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: b, Start state: 1, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: d, Start state: 1, End_state: 0  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: b, Start state: 1, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: d, Start state: 1, End_state: 0  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: a, Start state: 1, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: b, Start state: 2, End_state: 3  
Input symbol: b, Start state: 3, End_state: 3  
Input symbol: b, Start state: 3, End_state: 3  
Input symbol: b, Start state: 3, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: c, Start state: 1, End_state: 1  
Input symbol: c, Start state: 1, End_state: 1  
Input symbol: c, Start state: 1, End_state: 1  
Input symbol: d, Start state: 1, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: a, Start state: 0, End_state: 1  
Input symbol: a, Start state: 1, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: c, Start state: 0, End_state: 3  
Input symbol: c, Start state: 3, End_state: 1  
Input symbol: c, Start state: 1, End_state: 1  
Input symbol: c, Start state: 1, End_state: 1  
Input symbol: b, Start state: 1, End_state: 3  
Input symbol: b, Start state: 3, End_state: 3  
Input symbol: b, Start state: 3, End_state: 3  
Input symbol: b, Start state: 3, End_state: 3  
Input symbol: a, Start state: 3, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: a, Start state: 2, End_state: 2  
Input symbol: d, Start state: 2, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0  
Input symbol: d, Start state: 0, End_state: 0
```

3.

С использованием циклов:

```
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> gcc fibonacci_cycles.c -o fibonacci_cycles  
  
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> ./fibonacci_cycles  
x(1) = 1  
x(2) = 1  
x(3) = 2  
x(4) = 3  
x(5) = 5  
x(6) = 8  
x(7) = 13  
x(8) = 21  
x(9) = 34  
x(10) = 55
```

С использованием рекурсии:

```
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> gcc fibonacci_recursion.c -o fibonacci_recursion  
  
~/University/Innopolis/Programming for gamedeveloping/Homework/Homework 1 master*  
> ./fibonacci_recursion  
x(1) = 1  
x(2) = 1  
x(3) = 2  
x(4) = 3  
x(5) = 5  
x(6) = 8  
x(7) = 13  
x(8) = 21  
x(9) = 34  
x(10) = 55
```